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Robust Unsupervised Anomaly Detection With Variational Autoencoder in Multivariate Time Series Data

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ABSTRACT Accurate detection of anomalies in multivariate time series data has attracted much attention due to its importance in a wide range of applications. Since it is difficult to obtain accurately labeled data, many unsupervised anomaly detection algorithms for multivariate time series data have been developed. However, building such a system is challenging since it requires capturing temporal dependencies in each time series and must also encode the inter-correlations between different pairs of time series. To meet this challenge, we propose a Multi Scale Convolutional Variational Autoencoder (MSCVAE) to detect anomalies in multivariate time series data. Firstly, multi scale attribute matrices are constructed from multivariate time series to characterize multiple levels of the system states at different time steps. Then, given the attribute matrices, a convolutional variational autoencoder is employed to generate reconstructed attribute matrices, and also an attention-based ConvLSTM network is used to capture the temporal patterns. In addition, a new ERR-based threshold setting strategy is developed to optimize anomaly detection performance instead of relying on the traditional ROC-based threshold setting strategy with an imbalanced dataset. Finally, the proposed framework is assessed by means of experiments on four datasets. The experimental results show that our proposed framework is superior to competing algorithms in terms of model performance and robustness, demonstrating that our model is effective in detecting anomalies in multivariate time series.

INDEX TERMS Anomaly detection, multivariate time series, convolutional variational autoencoder, threshold setting strategy.

I. INTRODUCTION

In data mining, a time series [1] is a sequence of data points collected over time [2]. Such a sequence forms the basis of methods for tracking changes over time. Time series data can track changes in milliseconds, days, months, or even years. Time series data plays an important role in virtually all areas of science, engineering, commerce, and industry. Since data points in time series are collected at intervals, there is a relationship between successive observations, proportional or not, which distinguishes time series data from other kinds of data. There are two types of time series data. One is a univariate time series based on a single time dependent variable (or dimension). The other is a multivariate time series based on two or more time dependent, interrelated variables (or dimensions).

In recent years, research on time series data has focused on detecting and analyzing anomalies [3]–[6]. Anomaly detection is the identification of unexpected data points, i.e., events, or items that differ significantly from what is expected. Three types of time series anomalies have been identified, namely, point anomalies, contextual anomalies, and collective anomalies [7]. Point anomalies are points that exist far outside the range of the entire data set. Contextual anomalies are values that deviate significantly from most of the data points in the same context. Collective anomalies occur when a subset of the data points deviates substantially from the entire dataset. Anomaly detection is critical in many real world applications, such as analysis of potentially fraudulent transactions, sensor network faults, abnormal equipment behavior, etc.

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Great strides have been made recently in multivariate time series anomaly detection systems [8]–[11]. This progress is also reflected in many areas of application [12]–[16]. Multivariate time series anomaly detection refers to anomaly detection of time series data with multiple sequences. The occurrence of anomalies in multivariate time series data typically involves multiple features. Sequential analysis of individual features cannot accurately locate all the anomalies, because several variables have to be examined simultaneously when analyzing segments of data. Moreover, encoding the inter-correlations between different pairs of time series also needs to be considered. Clearly, detecting anomalous parts of multivariate time series is a challenging problem.

Machine learning techniques are increasingly being adopted to detect anomalies because they can capture different characteristics of time series and detect anomalies effectively. Various anomaly detection methods for multivariate time series data have been developed. One such method employs unsupervised anomaly detection based on a Convolutional encoder-decoder with an attention-based Convolutional Long-Short Term Memory (ConvLSTM) network to detect and diagnose anomalies from signature matrices [17]. Liang *et al.* [18] make use of the multi-channel signature matrix transform to capture hidden non-linear anomalies. They also employ a forgetting mechanism to assign different weights, and use a novel threshold strategy to optimize performance. Another method uses a fully convolutional adversarial autoencoder to detect anomalies in video scenes and localizations [19]. Yet another method applies online unsupervised anomaly detection to industrial robot performance; this approach is based on Variational Autoencoder and Convolutional neural network with Sliding window to better recognize normal patterns of data for real-time anomaly detection [20]. Despite considerable progress, VAE based anomaly detection for imbalanced data has not received much attention.

In this paper, we propose an unsupervised anomaly detection method, i.e., Multi-Scale Convolutional Variational Autoencoder (MSCVAE), to detect anomalies in multivariate time series data. This model incorporates a new threshold setting strategy designed to fill a gap in existing methods.

The main contributions of our proposed framework are as follows.

- The novel MSCVAE framework is designed to detect anomalies in multivariate time series data. MSCVAE constructs multi-scale attribute matrices to characterize multiple levels of the system states across different time steps and then uses a convolutional variational autoencoder to extract the characteristics of the time series input. Specifically, we use an attention-based Convolutional Long-Short Term Memory (ConvLSTM) network to capture the temporal patterns and also to reconstruct the attribute matrices.
- We propose a novel threshold setting strategy based on a confusion matrix to optimize threshold selection of anomaly detection, which will help to improve the model robustness under conditions of imbalance between normal and abnormal data in multivariate time series.
- Experiments have been conducted on four datasets in order to verify the effectiveness of the proposed framework and the new threshold setting strategy. The results demonstrate that our method is superior to competing models in terms of anomaly detection performance and robustness under different ratios of imbalanced datasets.

The paper is structured as follows. Section II analyzes related work. Section III describes the proposed methodology framework. The experiment is detailed in Section IV. Finally, conclusions are presented in Section V.

II. RELATED WORK

Anomaly detection is challenging, and many approaches have been taken in various applications. In past years, many classical unsupervised approaches have been developed [21]–[26], including Principal Component Analysis (PCA) [27], which finds a low-dimensional projection that captures most of the variance in the data. The anomaly score is the reconstruction error of this projection. It is a linear algebra technique that can automatically achieve dimension reduction. Lee *et al.* [28] proposed online over-sampling PCA which makes use of online platforms for large-scale problems. By over-sampling the minority class of the target instance, their proposed algorithm allows them to determine the anomaly of the target instance.

One of the latest techniques for dimensionality reduction is Autoencoder [29], which is a popular approach for anomaly detection. Autoencoders consists of an encoder and decoder which reconstruct data samples and use the reconstruction error as the anomaly score [30]. Zhou *et al.* [31] proposed a deep autoencoder that combines robust PCA and deep autoencoders. It splits data into two parts: one part can be reconstructed by autoencoders; the other is the noise (outliers) in the data. Deep Autoencoding Gaussian Mixture Model (DAGMM) [32] jointly considers the Deep Autoencoder and Gaussian mixture Model to model the density distribution of multi-dimensional data.

Recently, Generative Adversarial Networks (GANs) [33] and LSTM-based approaches [34] have also shown promising performance for multivariate anomaly detection [35], [36]. Multivariate Anomaly Detection for Time Series Data with Generative Adversarial Networks (MAD-GAN) [37] proposed an unsupervised anomaly detection method based on generative adversarial networks (GAN) by considering complex dependencies among different time series variables. The LSTM-based Encoder-Decoder [38] models time series temporal dependency by means of LSTM networks and achieves better generalization capability than traditional methods. OmniAnomaly [39] is a stochastic recurrent neural network designed to avoid potential misguiding by uncertain instances, which uses stochastic variable connection and normalizing flow to get reconstruction probabilities to determine anomalies. Ryota *et al.* [40] introduced a convolutional neural network and environment-dependent anomaly

detector to detect the object, its attributes, and actions in the image. An environment-specific model can identify unusual attributes likely to explain abnormal patterns. Hu *et al.* [41] proposed a time series anomaly detection technique using six meta-features. This technique is a One-class Support Vector Machine (OC-SVM) system designed to identify the abnormal states of a univariate or multivariate time series based on local dynamics. Recently, a novel computational approach, namely Local Recurrence Rate based Discord Search (LRRDS), was proposed to identify discords from multivariate time series. This approach reduces the dimensionality of a time series and can detect variable-length discords using the given time series as the normality reference [42].

U-Net [43] shares some of the design characteristics of the system architecture described in this paper. It is a fully convolutional neural network incorporating so-called skip channels between encoding and decoding layers. This approach allows for integrating high and low level features in a way that prevents information loss and reduces the depth of sequential layers. However, U-net is limited in that the rate of learning may be diminished in the middle layers of a high depth case. This means the system is at risk of ignoring layers with abstract features, thus limiting extraction of some of the complex features that could help image segmentation in medical images. Moreover, U-Net requires rather substantial training time because of a large number of hyper-parameters. By contrast, our system uses a more effective method consisting of the multi-scale MSCVAE model that has an attention-based ConvLSTM network. This method allows for capturing interesting features and key patterns to assist with anomaly detection in multivariate time series data.

Some Variational Autoencoders (VAEs) [44]–[48] have taken a probabilistic approach, and autoencoders have been combined with Gaussian mixture modeling [19]. Bayer and Osendorfer [49] used Variational Inference to learn the underlying distribution of sequences and introduced recurrent stochastic networks. The core of these models is an RNN extended with a latent variable. Sölch *et al.* [50] used a Stochastic Recurrent Network (STORN) to detect robot anomalies using unimodal signals. Park *et al.* [51] presented the combination of LSTM-VAE using multimodal sensory signals, where LSTM is used to replace the feed-forward network in VAE. Pereira *et al.* [52] proposed applying a self-attention mechanism to VAE to improve the encoding-decoding process for energy data. Despite the intrinsic unsupervised setting, most of them may still not be able to detect anomalies effectively since most of the methods cannot capture temporal dependencies across different time steps in multivariate time series data.

Our goal is to develop specific approaches for multivariate time series data, which take account of the correlation between the series. Multivariate time series data usually contain noise and mask the true anomalies. Also, we need to consider the characteristics of the data itself in addition to the correlation between series. Combining the advantages of the above models, the MSCVAE model we propose can not only capture the time pattern effectively but also has the ability to deal with noise. Although these methods use a variety of technologies and operate in different ways, common to all is the need to have indicators to judge anomalies.

In anomaly detection problems, thresholding (i.e., setting a threshold) is an effective strategy for evaluating anomaly scores. The scores output by the model are separated into outliers and normal data using the threshold. Most systems select the threshold with the aid of the receiver operating characteristics (ROC) curve. The ROC curve is a plot in a unit square of true positive rate (TPR) versus false positive rate (FPR). Using the ROC curve, the threshold that optimizes specificity as well as sensitivity can be identified. However, the imbalanced class distribution of a data set poses a problem because most unsupervised learning techniques are designed for balanced class distribution [53]. To solve this problem, we propose a new threshold setting strategy based on a confusion matrix, thus selecting a better threshold, improving model performance and robustness.

III. OVERVIEW OF THE METHODOLOGY

In this section, we first introduce the problem to be tackled in section III-A, and then elaborate on how to generate attribute matrices in section III-B. Next, we explain the proposed Multi-scale convolutional variational autoencoder (MSCVAE) in detail in section III-C - III-G. Specifically, we first show how to generate multi-scale system attribute matrices. Then, we encode the spatial information in attribute matrices via a convolutional variational encoder and model the temporal information via attention-based ConvLSTM. We reconstruct attribute matrices based upon a convolutional variational decoder. Finally, we detail the strategy of threshold setting in section III-H. Figure 1 presents an overview of the MSCVAE framework. The terminology and notation used in the paper are summarized in Table 1.

A. PROBLEM STATEMENT

In this work, we focus on multivariate time series, given the historical data of n time series $X = (x_1, x_2, \ldots, x_n)^T \in \mathbb{R}^{n \times T}$, of length T, and assume that there are anomalies in the data. We aim to detect anomalous events at certain time steps after T. We use only the normal dataset for training to characterize the various time series patterns under normal conditions. For the validation, we also use only the normal dataset. Both normal and abnormal data are used for testing.

B. GENERATE ATTRIBUTE MATRICES

Given the importance of correlations between the different pairs of time series for characterizing the system state [54], we generate an $n \times n$ attribute matrix M^t utilizing the pairwise inner product of multivariate time series within this segment. This is designed to illustrate the inter-correlations between different pairs of time series in a multivariate time series segment from time $t - w$ to t. We adopt the method for calculating the attribute matrix proposed in [17] to capture

TABLE 1. Terminology and notation used in the paper.

Notations	Descriptions
X	Multivariate time series dataset
T	Length of time series
n	Dimension of time series
w	Size of sliding window
t	Time step
M^t	Attribute matrix
$p(z)$	Prior distribution
$p(z\|x)$	Probabilistic encoder
$p(\mathrm{x}\|z)$	Probabilistic decoder
$P^{t,0}$	Input of the first layer of the encoder
$N(0,I)$	A multivariate unit Gaussian distribution
z	Latent variable
$*$	Convolutional operation
o	Hadamard product
$\#$	Deconvolutional operation
$\oplus$	Concatenation operation
$f(\cdot)$	Activation function unit
$\hat{P}^{t,0}$	Output, reconstruction at t time step
$i^{t,l}$, $f^{t,l}$, $o^{t,l}$	3D tensor
$G^{t,l}$	Candidate memory
$C^{t,l}$	Cell state
$H^{t,l}$	Current hidden state

TABLE 2. The pseudocode of attribute matrices generation.

Algorithm Generating attribute matrices

Input: Multivariate time series X, length of time series T, window size w, dimension of time series n

Output: Attribute matrix m

1: **for** i in T **do**
2: for j in T **do**
3: $X_i = x[t-w+1], x[t-w+2], \ldots, x[t]$
4: $X_j = x[t-w+1], x[t-w+2], \ldots, x[t]$
5: $m[i][j] = \frac{X_i \times X_j}{w}$
6: return M

the similarity of shape and value scale correlations between two time series. Examples of attribute matrices are shown in Figure 1, part A. The pseudocode of the algorithm for generating the attribute matrix is introduced in Table 2. For multivariate time series segment X^w, we define two time series, namely, $x_i = \left\{x_i^{t-w}, x_i^{t-w+1}, \ldots, x_i^t\right\}$, $x_j = \left\{x_j^{t-w}, x_j^{t-w+1}, \ldots, x_j^t\right\}$, and their correlation $m_{ij}^t \in M^t$ can be computed as follows:

$$m_{ij}^t = \frac{\sum_{\gamma=0}^{w} x_i^{t-\gamma} x_j^{t-\gamma}}{w} \quad (1)$$

where w is the sliding window size, assigned the value 10 at each time step. Also, the interval between two segments is set to 10. To further explain the attribute matrices generation process, the algorithm's pseudocode is shown in Table 2.

C. VARIATIONAL AUTOENCODER (VAE)

Autoencoder is a typical neural network that transforms inputs into outputs with the minimum possible error. It consists of three sequential layers: the latent variable layer, the encoder layer, and the decoder layer. The encoder encodes or compresses the input data $x \in \mathbb{R}^{d_x}$ into a latent variable $z \in \mathbb{R}^{d_z}$, while the decoder decodes or reconstructs the encoded data (latent variable) z back to the input dimension. The objective of Autoencoder is to get a reconstructed value from the latent variable that is close to the original input data. A disadvantage of the autoencoder is that a hidden layer may not be continuous and poses serious computational challenges. Variational Autoencoder (VAE) is a type of Autoencoder designed to address this issue.

VAE is an unsupervised learning network for dimension reduction, which can learn to represent complex data without supervision using a deep neural network. VAE relies on probability distributions of observations in latent variables and makes strong assumptions about the distribution of latent variables. Therefore, instead of generating an encoder that outputs a single value to describe each latent variable, it creates an encoder that operates with a probability distribution for each latent variable.

VAE has three main layers, namely, encoder, decoder, and loss function. The input denoted by x is generated from a latent variable z. VAE defines latent variable z as a random variable distributed according to a prior function $p(z)$, defined as a multivariate unit Gaussian distribution $N(0, I)$. The first process samples data x from the conditional likelihood distribution $p(x|z)$. Then, the decoder $p(x|z)$ functions as a generator model corresponding to the distribution of the decoded variable given the encoded one. However, the encoder is defined by the conditional probability function $p(z|x)$, i.e., the distribution of the encoded variable given the decoded one.

The loss function of VAE has two terms: the first term is a reconstruction loss term, which can maximize the reconstruction likelihood, and the second term is a regularization term, which ensures that the distribution learning $q(z|x)$ is like the true prior distribution $p(z)$. The loss function is given by Eq. 2.

$$L(x_i) = -E_{z\sim q(z|x_i)}[\log p(x_i|z)] + KL(q(z|x_i) \parallel p(z)) \quad (2)$$

The regularization term of VAE is the Kullback-Leibler divergence which measures the difference between the encoder's distribution $q(z|x)$ and $p(z)$, and constitutes a measure of how close p is to q, as described in Figure 1. part B.(a).

D. CONVOLUTIONAL ENCODER

We use convolution encoders [55] to encode the spatial form of the system attribute matrices. Four convolutional encoder layers are applied to extract the values of the attribute matrices in our framework. $P^{t,0}$ represents the input of the first

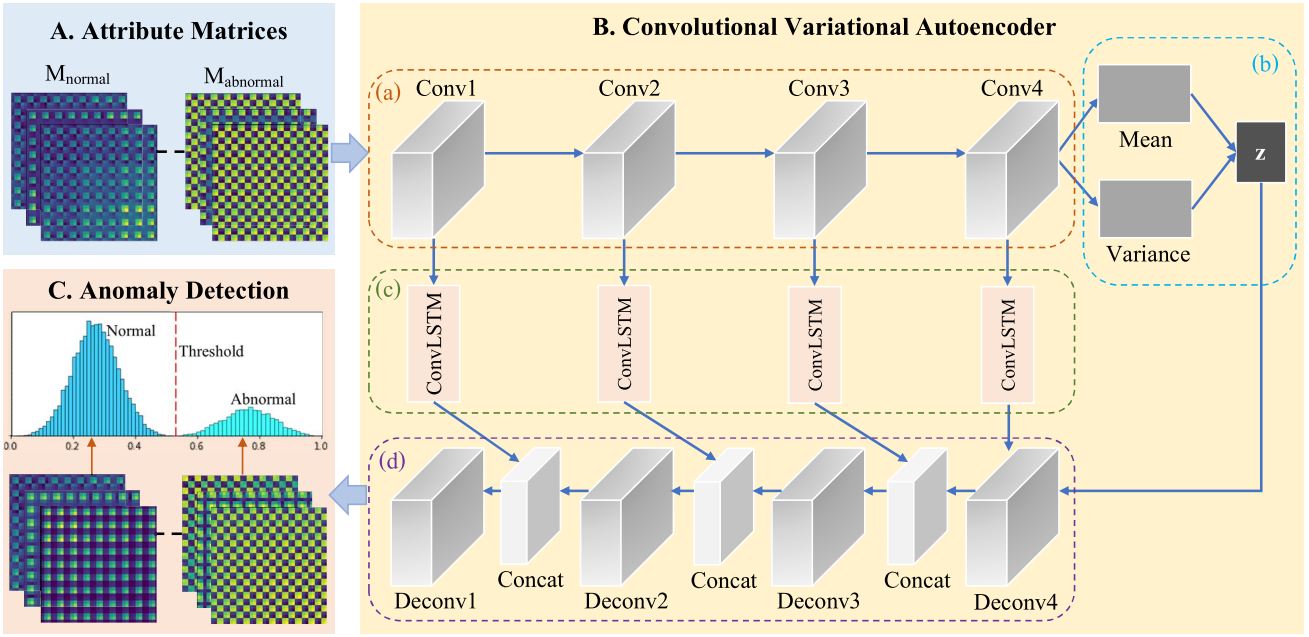


FIGURE 1. Concept of the MSCVAE model: A. Generate attribute matrices. B. (a) Attribute matrices encoding via convolutional encoders. B. (b) VAE learns mean coding and standard deviation coding and then samples randomly from a gaussian distribution to generate the latent variables z. B. (c) Temporal patterns are modeled by attention-based convolutional LSTM networks. B.(d) Latent variables are decoded to reconstruct the input via deconvolutional decoders. C. anomaly detection.

layer and assumes that $P^{t,\,l-1} \in R^{n_{l-1}\times n_{l-1}\times d_{l-1}}$ denotes the feature maps in $(l-1)-$ th layer. The output of $l-$ th layer is given by:

$$P^{t,\,l} = f(W_e^{\,l} * P^{t,\,l-1} + b_e^l) \tag{3}$$

where $*$ denotes the convolutional operation, $W_e^{\,l}$ is the filter kernel of layer l, b_e^l is a term of bias, $P^{t,\,l}$ is the output of layer l, $P^{t,\,l-1}$ is the output of the $l-1$ layer (input of layer l), and $f(\cdot)$ is an activation function. Figure 1. part B.(b) illustrates the detailed encoding process of attribute matrices.

E. ATTENTION-BASED CONVOLUTIONAL LSTM

The spatial feature maps, extracted by the convolutional encoder, depend on the previous time steps. The traditional Convolutional LSTM performance might degrade as the length of the sequence increases. Xingjian *et al.* [56] introduced the LSTM model to the ConvLSTM to capture the temporal information in a video sequence. Here we have used an attention-based ConvLSTM to obtain the temporal characteristics which can be adaptive with different time steps. ConvLSTM consists of four convolutional layers of four ConvLSTM blocks. Details of ConvLSTM are shown in Figure 1. part B.(c). At time t, the results of $P^{t,\,l}$ calculated by Eq. 3. in layer l of the convolutional encoder are used as input to the $l-$ th layer of ConvLSTM. The other input is the time $t-1$ hidden state $H^{\,t-1,\,l}$ in hidden layer l. The ConvLSTM cell is formulated as:

$$\begin{aligned}
i^{t,l} &= \sigma(W_{pi}^l * P^{t,l} + W_{hi}^l * H^{t-1,l} + W_{ci}^l \circ C^{t-1,l} + b_i^l)\\
f^{t,l} &= \sigma(W_{pf}^l * P^{t,l} + W_{hf}^l * H^{t-1,l} + W_{cf}^l \circ C^{t-1,l} + b_f^l)\\
o^{t,l} &= \sigma(W_{po}^l * P^{t,l} + W_{ho}^l * H^{t-1,l} + W_{co}^l \circ C^{t-1,l} + b_o^l)\\
G^{t,l} &= \tanh(W_{pc}^l * P^{t,l} + W_{hc}^l * H^{t-1,l} + b_c^l)\\
C^{t,l} &= f^{t,l} \circ C^{t-1,l} + i^{t,l} \circ G^{t,l}\\
H^{t,l} &= o^{t,l} \circ \tanh(C^{t,l})
\end{aligned} \tag{4}$$

where σ is the sigmoid function, $W_{pi}^{\,l}$ is the filter kernel of the input gate, $W_{ci}^{\,l}$ is the filter kernel of the input gate process for the input of the hidden layer at the previous time step, and $W_{hi}^{\,l}$ is the filter kernel of the cell state at the previous time step in the input gate. The input $P^{t,\,l}$, the cell state $C^{t,\,l}$, the hidden state $H^{t,\,l}$, the candidate memory $G^{t,\,l}$, and the gates $i^{t,\,l}$, $f^{\,t,\,l}$, $o^{\,t,\,l}$ are all 3D tensors. The symbol "$*$" denotes the convolutional operator, and "$\circ$" denotes the Hadamard product. Figure 1. part B.(c) illustrates the temporal modeling procedure.

We follow the idea of the temporal attention mechanism mentioned in [17], which is given by:

$$\hat{H}^{t,l} = \sum_{i\epsilon(t-h,t)} \alpha^i H^{i,l} \tag{5}$$

$$\alpha^i = \frac{\exp(s^t)}{\sum_{i\epsilon(t-h,t)} \exp(s^t)} \tag{6}$$

$$s^t = \frac{(H^{t,l})^T (H^{i,l})}{\chi} \tag{7}$$

where χ is the rescale factor ($\chi = 5$). That is, we take the last hidden state $H^{t,l}$ as the group level context vector and measure the importance weights α^i of previous steps through a softmax function.

F. CONVOLUTIONAL DECODER

We can reconstruct the input data using the extracted attribute output produced by the decoder. For the convolutional decoder, we follow the procedure discussed in [17] which is formulated as:

$$\hat{P}^{t,\,l-1} = \begin{cases} f(\hat{W}_d^{t,\,l}\hat{H}^{t,\,l} + \hat{b}_d^{t,\,l}), & l = 4 \\ f(\hat{W}_d^{t,\,l}[\hat{H}^{t,\,l} \oplus \hat{P}^{t,\,l}] + \hat{b}_d^{t,\,l}), & l = 3,\ 2,\ 1 \end{cases} \tag{8}$$

where $\circledast$ denotes the deconvolutional operation, $\oplus$ is the concatenation operation, $f(\cdot)$ is the activation function unit (same as in the encoder). $\hat{W}_d^{t,\,l}$ and $\hat{b}_d^{t,\,l}$ are the filter kernel and bias parameter, respectively, of the $l-$th deconvolutional layer. We reconstruct the attribute matrix of each layer by reversely decoding the deconvolution from layer $l = 4$ to layer $l = 1$. We then deconvolve the result $o^{t,\,4}$ of the ConvLSTM at the last layer $l = 4$ and reconstruct the previous layer matrix $\hat{P}^{t,\,3}$. After that, the deconvolution operation is performed as concatenation to reconstruct the attribute matrix sequence of the $l-1-$th layer. Figure 1. part B.(d) illustrates the decoding procedure.

G. LOSS FUNCTION

Mean Square Error (MSE) is used as a loss function to optimize the parameter values in our model. MSE is the most commonly used loss function suitable for anomaly detection problems. MSE is given by the following:

$$L_{MSCVAE} = \sum_{t=1}^{k}\sum_{c=1}^{k}\left\|P^{t,\,0} - \hat{P}^{t,\,0}\right\|^2 \tag{9}$$

where $P^{t,0} \in \mathbb{R}^{n\times n}$. We employ the mini-batch stochastic gradient descent method and the Adam optimizer to minimize the above loss. After sufficient training epochs, the learned neural network parameters are utilized to infer the reconstructed attribute matrices of validation and test data. Finally, we perform anomaly detection and diagnosis based on the residual attribute matrices, elaborated in the next section.

H. THRESHOLD SETTING STRATEGY

VAEs map each group of attribute matrices to anomaly scores during the anomaly detection process. The threshold setting aims to give the best boundary to distinguish normal and abnormal samples, and then detected samples are labeled normal or abnormal by means of VAEs.

The confusion matrix is commonly used to show how each test value predicts classes compared to their actual classes including True Positive (TP), False Positive (FP), True Negative (TN), and False Negative (FN).

Based on the ROC-based threshold setting strategy [57], the best threshold is always sought as the dot on the upper left corner or point (0,1). However, the ROC-based threshold setting strategy is insensitive to imbalanced datasets, leading to poor performance in anomaly detection. According to the false positive rate formula, when a large number of normal samples are wrongly judged as anomalies, the false positive rate (FPR) changes a little. Therefore, anomaly detection performance may be widely divergent for two threshold values that are close together. In this case, it is not easy to select an optimal threshold among different discrete points. Moreover, Eq. 11 is often adopted to calculate the distance between a point on the ROC curve and the point (0,1). The threshold corresponding to the minimum distance is selected as the best threshold. However, TN is much larger than TP, which makes $\frac{FP^2}{FP^2+TN^2}$ much smaller than $\frac{FN^2}{FN^2+TP^2}$. Therefore, when choosing the minimum distance, the ROC-based strategy focuses more on the former, only choosing the smaller FN as much as possible. Eventually, it leads to low F1.

$$\text{Distance} = \sqrt{(1-TPR)^2 + FPR^2} \tag{10}$$

$$= \frac{FN^2}{FN^2+TP^2} + \frac{FP^2}{FP^2+TN^2} \tag{11}$$

We adopt a threshold setting strategy using a confusion matrix to avoid the problems described above. The confusion matrix can effectively reflect the anomaly detection results, even for an imbalanced dataset. Therefore, we introduce a new error rate (ERR) defined as a function of TP, FP, FN, and TN, as shown in the formula below. The aim of the new threshold setting strategy is to minimize ERR, which means fewer samples are misjudged. As a result, the optimal threshold can be selected according to the minimum ERR.

$$\text{ERR} = \frac{FP}{FP+TP+TN} \tag{12}$$

As indicated above, Precision, Recall, F1-Score, and ERR are utilized as evaluation metrics for anomaly detection. Moreover, the geometric mean or G-mean [58] of sensitivity and specificity, is suitable for evaluating the quality of binary (two-class) classifications for a balanced as well as imbalanced dataset problem. Therefore, G-mean is used as another evaluation metric in the following experiments.

$$\text{G-mean} = \sqrt{\text{Recall} * \text{Specificity}} \tag{13}$$

$$= \sqrt{\frac{TP}{TP+FN} \times \frac{TN}{TN+FP}} \tag{14}$$

IV. EXPERIMENTS AND DISCUSSION

A. DATASETS DESCRIPTION

To illustrate the performance of the proposed method, we have carried out extensive experiments on four publicly available benchmark datasets, namely, Satellite, Wafer (UCR), EEG, and Opt. Descriptions of these data sets are given below, and the details of the experiments are listed in Table 3. The second dataset can be downloaded from the website: http://www.cs.ucr.edu/∼ eamonn/discords/, and the other three datasets are from the UCI public database. Details of the above datasets are given below.

1) SATELLITE

This dataset was generated from data purchased from NASA by the Australian Centre for Remote Sensing. It consists of the multi-spectral values of pixels in 3×3 neighborhoods in a

satellite image together with the classification associated with the central pixel in each neighborhood. The data set is used to predict the category of the image of the observed region of soil. Soil classes ''red soil'', ''grey soil'', ''mixture class'', and ''very damp grey soil'' constitute the normal class. The class of anomalies consists of ''cotton crop'', ''damp grey soil'', and ''soil with vegetation stubble''. This classification is based on multi-spectral values, consisting of 36 time series and 6435 instances.

2) WAFER

The Wafer dataset is related to semiconductor microelectronics fabrication, using data collected from various sensors during the processing of silicon wafers for semiconductor fabrication. Each time series in this dataset contains measurements recorded by one sensor in the course of processing one wafer by one tool. The dataset contains 152 attributes, and there is a large class imbalance between normal and anomaly.

3) EEG

This dataset is from one continuous EEG measurement with the Emotiv EEG Neuroheadset. The duration of the measurement was 117 seconds. The eye state was detected via a camera during the EEG measurement and later added manually to the file after analyzing the video frames. This dataset consists of 14 EEG attribute values and one indicating the eye state.

4) OPT

The Opt dataset is used for preprocessing programs made available by NIST to extract normalized bitmaps of handwritten digits from a preprinted form, which contains 64 parameters. It is a character recognition dataset for integers 0-9. The 32×32 bitmaps are divided into non-overlapping

blocks of 4×4 each, which generates an 8×8 input matrix. The instances of digits 1-9 are treated as inliers whereas the instances of the digit 0 are treated as outliers.

B. EXPERIMENTAL SETUP

Our experiments use the open-source machine learning library Scikit-learn, and deep learning framework Torch, Keras, and TensorFlow to develop the baseline models and our framework. The computer configuration is Intel(R) Core(TM) i7-9750H CPU 2.60GHz 6-Core Processor with NVIDIA GeForce RTX 2060 GPU, 16G Memory.

Furthermore, seventeen algorithms were used to verify the superiority of the proposed framework, including classical anomaly detection algorithms and deep architecture models.

The classical anomaly detection algorithms include K-Nearest Neighbor (KNN), Local Outlier Factor (LOF), Isolation Forest (iForest), One-class Support Vector Machine (OCSVM), Gaussian Mixture Model (GMM), Principal Component Analysis (PCA), Angle-based Outlier Detector (ABOD), Histogram-based Outlier Score (HBOS), Cluster-based Local Outlier Factor (CBLOF), Multiple-Objective Generative Adversarial Active Learning (MOGAAL).

TABLE 3. The detailed information of time series data sets.

Datasets	#Training	#Validation	#Testing	Length	Anomaly rate
Satellite	3,000	1,000	2,435	36	31.6 %
Wafer	4,000	1,000	2,164	152	10.6 %
EEG	8,000	2,000	4,980	14	44.9 %
Opt	2,500	1,000	1,716	64	2.9 %

TABLE 4. Detailed structural comparison of the algorithms.

Model	Kernel	Dense	CNN	LSTM	Dropout
KNN	×	×	×	×	×
LOF	×	×	×	×	×
iForest	×	×	×	×	×
OCSVM	○	×	×	×	×
GMM	×	×	×	×	×
PCA	×	×	×	×	×
ABOD	×	×	×	×	×
HBOS	×	×	×	×	×
CBLOF	×	×	×	×	×
MOGAAL	×	×	×	×	×
DAGMM	×	○	×	×	○
GAN	×	○	×	×	×
MAD-GAN	×	○	×	○	○
OmniAnomaly	×	○	×	×	×
Autoencoder	×	○	×	×	×
VAEs	×	○	×	×	×
LSTM-AE	×	○	×	○	○
MSCVAE	×	×	○	○	×

The deep architecture models include Deep Autoencoding Gaussian Mixture Model (DAGMM), Generative Adversarial Network (GAN), Multivariate Anomaly Detection with GAN (MAD-GAN), OmniAnomaly, Autoencoder, Variational Autoencoder (VAEs), and Long Short-Term Memory Autoencoder (LSTM-AE). Table 4 presents the details of all the algorithms.

C. PERFORMANCE EVALUATION

All anomaly detection models in experiments were trained with the corresponding training subsets, consisting of normal samples. Then the models were verified using a validation method consisting of normal samples and testing subsets, including both normal and abnormal data. The model evaluation metrics, i.e., precision, recall, F1-Score, and G-mean, were in the range of 0 to 1. Higher precision, recall, F1-Score, G-mean, and lower ERR indicate better model performance. Table 5, Table 6, Table 7, and Table 8 show the confusion matrix elements and evaluation metrics of anomaly detection models on four datasets. The bold fonts in Table 5, Table 6, Table 7, and Table 8 indicate that the MSCVAE is superior to the other models.

The comparison experiments show that 10 classical anomaly detection algorithms and 7 deep architecture models have been implemented. Generally speaking, the ability

TABLE 5. Anomaly detection results of satellite dataset.

Model	Confusion matrix value				Performance evaluation				
	TP	FP	FN	TN	ERR	Precision	Recall	F1-Score	G-mean
KNN	45	75	68	51	0.4386	0.3750	0.3982	0.3863	0.4015
LOF	54	66	56	63	0.3607	0.4500	0.4909	0.4696	0.4896
iForest	51	69	58	61	0.3812	0.4250	0.4679	0.4454	0.4686
OCSVM	55	65	70	49	0.3846	0.4583	0.4400	0.4490	0.4349
GMM	72	48	73	46	0.2892	0.6000	0.4966	0.5434	0.4929
PCA	112	8	11	108	0.0351	0.9333	0.9106	0.9218	0.9207
ABOD	44	76	73	46	0.4578	0.3667	0.3761	0.3713	0.3766
HBOS	48	72	70	49	0.4260	0.4000	0.4068	0.4034	0.4059
CBLOF	51	69	62	57	0.3898	0.4250	0.4513	0.4378	0.4519
MOGAAL	52	68	78	41	0.4224	0.4333	0.4000	0.4160	0.3879
DAGMM	80	40	95	24	0.2778	0.6667	0.4571	0.5424	0.4140
GAN	85	37	78	39	0.2298	0.6967	0.5215	0.5965	0.5173
MAD-GAN	82	37	78	42	0.2298	0.6891	0.5125	0.5878	0.5220
OmniAnomaly	84	37	79	39	0.2313	0.6942	0.5153	0.5915	0.5142
Autoencoder	83	37	78	41	0.2298	0.6917	0.5155	0.5907	0.5206
VAEs	105	9	31	94	0.0433	0.9211	0.7721	0.8400	0.8394
LSTM-AE	105	11	70	53	0.0651	0.9052	0.6000	0.7216	0.7049
MSCVAE	122	4	6	107	**0.0172**	**0.9683**	**0.9531**	**0.9606**	**0.9585**

TABLE 6. Anomaly detection results of wafer dataset.

Model	Confusion matrix value				Performance evaluation				
	TP	FP	FN	TN	ERR	Precision	Recall	F1-Score	G-mean
KNN	58	48	37	67	0.2775	0.5472	0.6105	0.5771	0.5964
LOF	60	46	56	48	0.2987	0.5660	0.5172	0.5405	0.5139
iForest	54	52	43	61	0.3114	0.5094	0.5567	0.5320	0.5482
OCSVM	61	45	56	48	0.2922	0.5755	0.5214	0.5471	0.5187
GMM	60	46	45	59	0.2788	0.5660	0.5714	0.5687	0.5666
PCA	61	45	42	62	0.2679	0.5755	0.5922	0.5837	0.5858
ABOD	50	50	45	65	0.3030	0.5000	0.5263	0.5128	0.5454
HBOS	49	57	41	63	0.3373	0.7075	0.5034	0.5882	0.5346
CBLOF	33	73	62	42	0.4932	0.4623	0.5444	0.5000	0.3562
MOGAAL	45	61	50	54	0.3813	0.3113	0.3474	0.3284	0.4716
DAGMM	75	31	74	30	0.2279	0.4245	0.4737	0.4478	0.4975
GAN	120	7	15	68	0.1170	0.9449	0.8889	0.9160	0.8977
MAD-GAN	112	12	21	65	0.1864	0.9032	0.8421	0.8716	0.8431
OmniAnomaly	107	10	33	60	0.2575	0.9145	0.7643	0.8327	0.8094
Autoencoder	100	12	49	49	0.0745	0.8929	0.6711	0.7663	0.7342
VAEs	90	21	12	87	0.1061	0.8108	0.8824	0.8451	0.8431
LSTM-AE	86	9	64	51	0.0616	0.9053	0.5733	0.7020	0.6981
MSCVAE	142	2	0	66	**0.0095**	**0.9861**	**1.0000**	**0.9930**	**0.9852**

of classical anomaly detection algorithms is limited when facing modeling issues for multivariate time series, and this conclusion is supported by results for the two comprehensive indexes, F1 and G-mean. DAGMM reduces the dimension and extracts features by using neural networks, slightly improving compared to GMM. GAN achieves similar results to DAGMM on the Satellite dataset (shown in Table 5) but performs well on the other three datasets. MAD-GAN attempts to map data into the latent space and detect anomalies via discriminant results and reconstruction errors generated from the mapping process. MAD-GAN achieves good results on the Opt dataset and gives slightly better results than OmniAnomaly. LSTM-AE performs better on low dimension datasets than high dimension datasets (shown in Tables 6 and 8). As an excellent sequence to sequence model, Autoencoder fails at temporal data and performs well on the Opt dataset as shown in Table 8. VAEs achieve better results than Autoencoder on four datasets, which suggests that a high dimension dataset is a challenge for the training of VAEs.

MSCVAE is superior to the other algorithms discussed above, except in the EEG dataset, which gives slightly better recall, F1-Score, and G-mean values than our proposed framework. Also, the Opt dataset gives slightly better recall values than in our proposed framework. Nevertheless, it is clear that our proposed framework can effectively improve

TABLE 7. Anomaly detection results of EEG dataset.

Model	Confusion matrix value				Performance evaluation				
	TP	FP	FN	TN	ERR	Precision	Recall	F1-Score	G-mean
KNN	164	92	94	140	0.2323	0.6406	0.6357	0.6381	0.6193
LOF	152	104	123	111	0.2834	0.5938	0.5527	0.5725	0.5342
iForest	184	72	85	149	0.1778	0.7188	0.6840	0.7010	0.6791
OCSVM	163	93	96	138	0.2360	0.6367	0.6293	0.6330	0.6132
GMM	170	86	113	121	0.2281	0.6641	0.6007	0.6308	0.5926
PCA	189	67	76	158	0.1618	0.7383	0.7132	0.7255	0.7077
ABOD	160	96	111	123	0.2533	0.6250	0.5904	0.6072	0.5758
HBOS	178	78	77	157	0.1889	0.6953	0.6980	0.6967	0.6829
CBLOF	157	99	103	131	0.2558	0.6133	0.6038	0.6085	0.5865
MOGAAL	92	164	146	88	0.4767	0.3594	0.3866	0.3725	0.3674
DAGMM	194	62	173	61	0.1956	0.7578	0.5286	0.6228	0.5120
GAN	297	10	33	150	0.0219	0.9674	0.9000	**0.9325**	0.9186
MAD-GAN	297	26	27	140	0.0562	0.9195	0.9167	0.9181	0.8793
OmniAnomaly	296	14	38	142	0.0310	0.9548	0.8862	0.9193	0.8982
Autoencoder	225	34	142	89	0.0977	0.8687	0.6131	0.7188	0.6660
VAEs	259	16	23	192	0.0343	0.9418	**0.9184**	0.9300	**0.9208**
LSTM-AE	251	14	117	108	0.0375	0.9472	0.6821	0.7930	0.7770
MSCVAE	296	9	34	151	**0.0197**	**0.9705**	0.8970	0.9323	0.9201

anomaly detection performance on multivariate time series. In other words, MSCVAE is much better than baseline methods, as it can handle both inter-sensor correlations and temporal patterns of multivariate time series effectively.

The reasons for the superior performance of our framework can be summarized as follows: (1) As a deep learning model with temporal patterns, MSCVAE can achieve good performance with no need for supervised training. (2) VAEs with an attention-based ConvLSTM network framework can effectively identify anomalies. (3) Our framework can model both inter-sensor correlations and temporal patterns of multivariate time series effectively.

D. ABLATION STUDY

We have conducted an extensive study to illustrate the impact of different components on the model results, using two key modules of our model: multi-scale attribute matrices and an attention-based ConvLSTM. Three variants of MSCVAE are considered in the evaluation:

- MSCVAE_w: MSCVAE framework without attention-based ConvLSTM.
- CVAE_a: MSCVAE framework without the multi-scale attribute matrices.
- CVAE_w: MSCVAE framework without both multi-scale attribute matrices and an attention-based ConvLSTM.

F1-Scores and G-means on four datasets are reported in Fig. 2. We observe that our proposed framework, MSCVAE (marked in blue), is obviously superior to the other three competing models on anomaly detection tasks, which indicates the importance of multi-scale attribute matrices and an attention-based ConvLSTM. However, MSCVAE_w (marked in pink) and CVAE_w (marked in yellow) methods can obtain approximately equal results with all four datasets. Besides, the results of CVAE_a (marked in green) are much better than

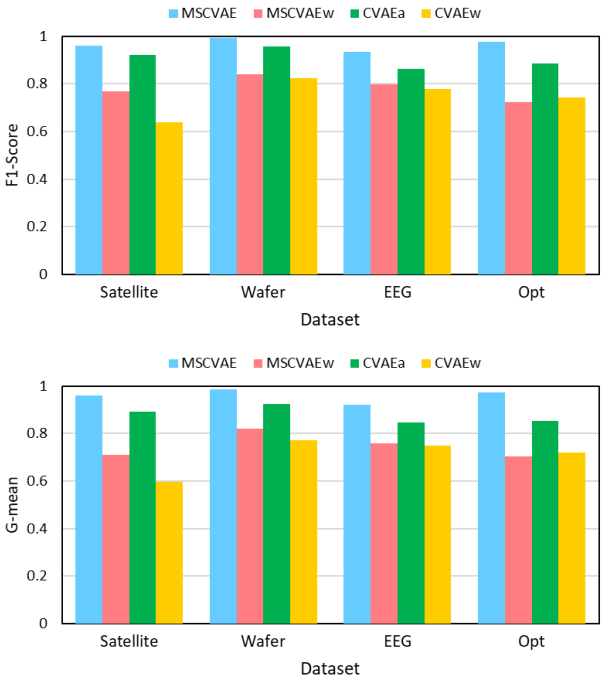
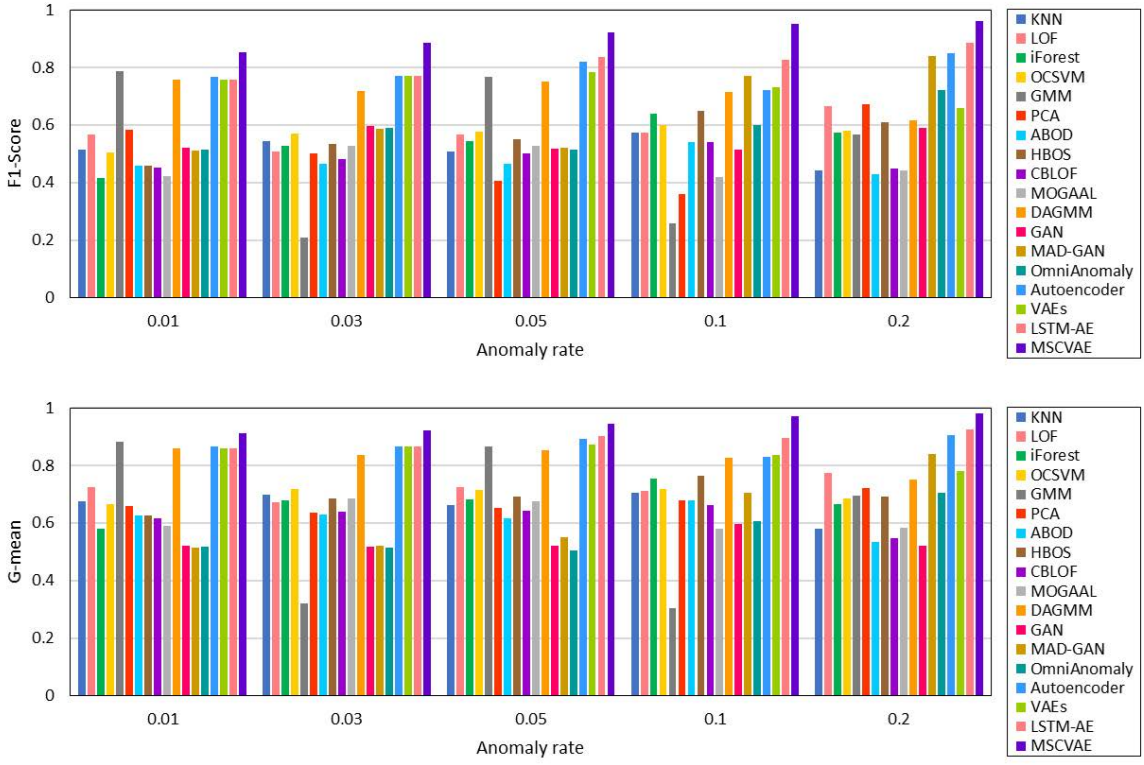


FIGURE 2. F1-Score and G-mean comparison of all four datasets under three different competing models.

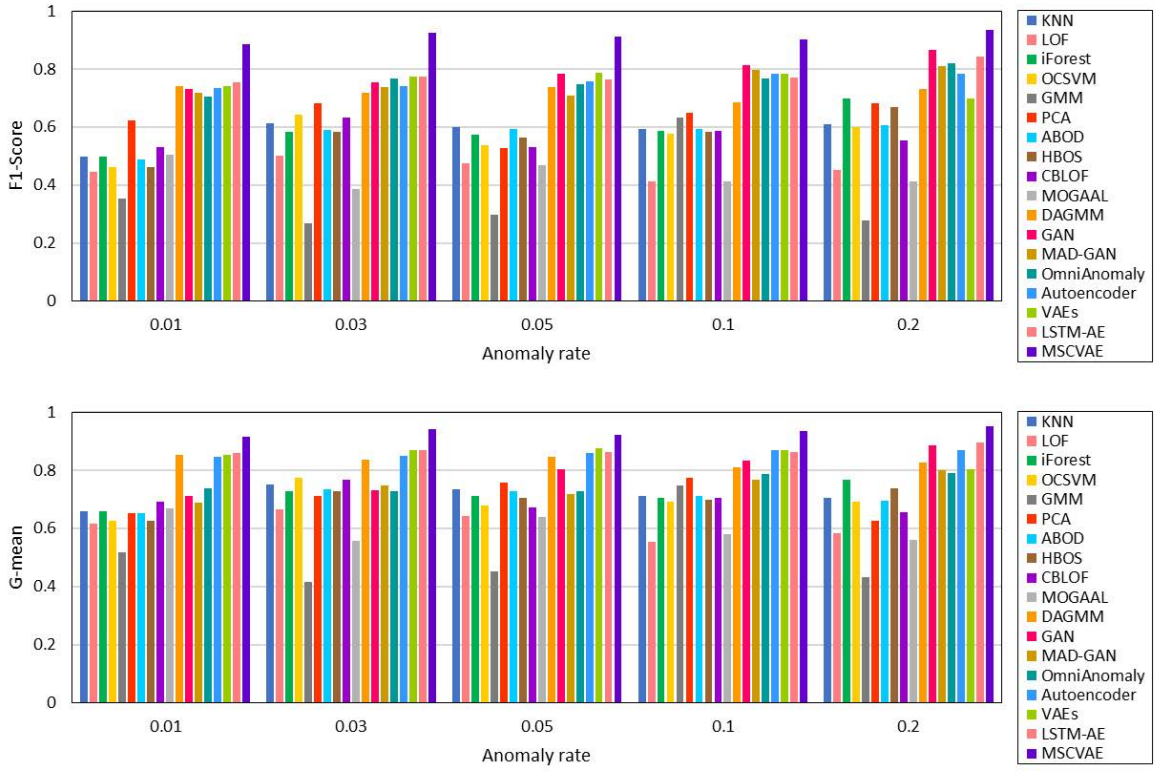
MSCVAE_w and CVAE_w, which indicates the importance of an attention-based ConvLSTM. Furthermore, we see that under all conditions, both multi-scale attribute matrices and attention-based ConvLSTM can effectively improve the anomaly detection performance of multivariate time series.

E. ROBUSTNESS EVALUATION

Anomaly detection often suffers from the dataset imbalance problem, which means there are more normal samples than

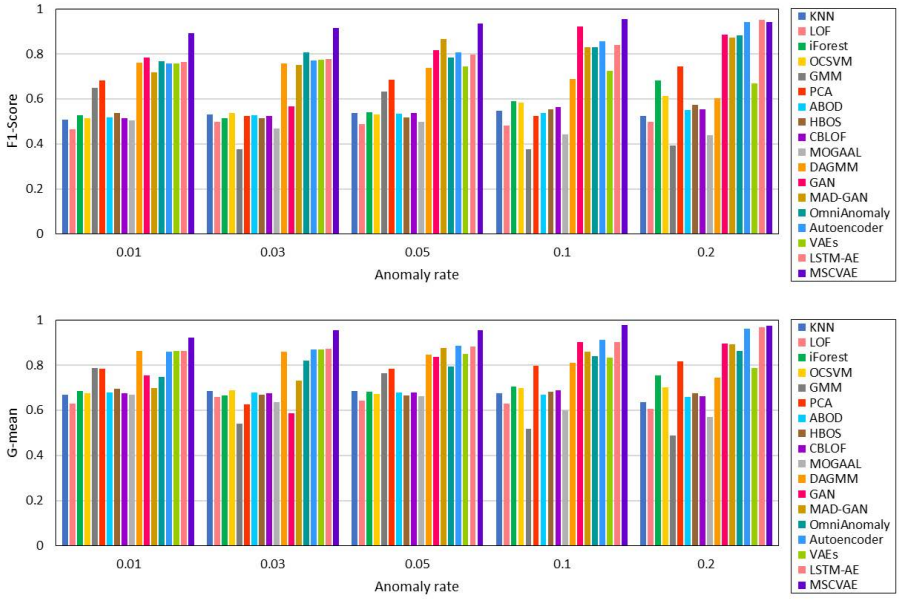


(a) Satellite dataset

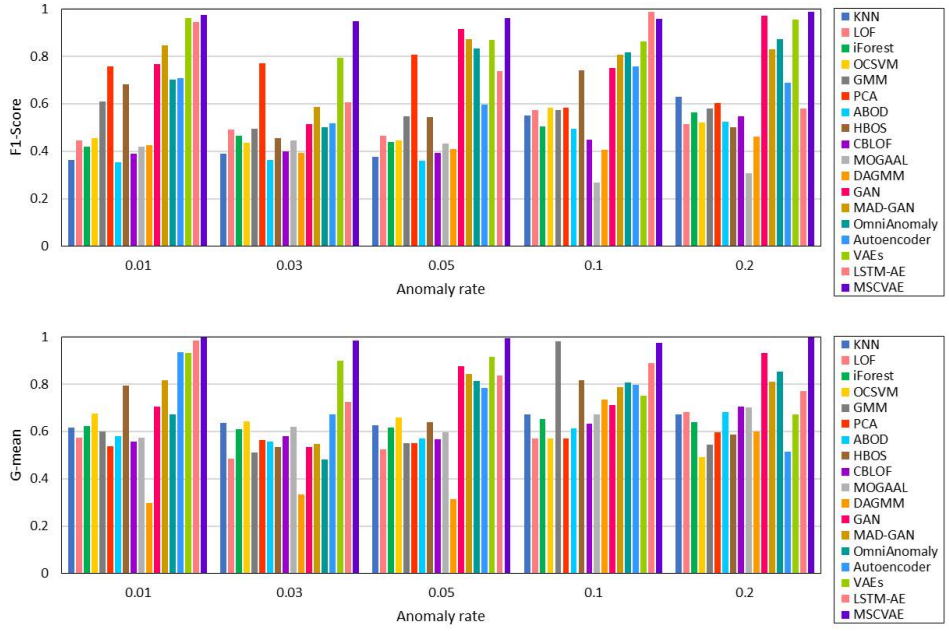


(b) Wafer dataset

FIGURE 3. F1-Score and G-mean comparison all four datasets under different anomaly rate.



(c) EEG dataset



(d) Opt dataset

FIGURE 3. ***(Continued.)*** **F1-Score and G-mean comparison all four datasets under different anomaly rate.**

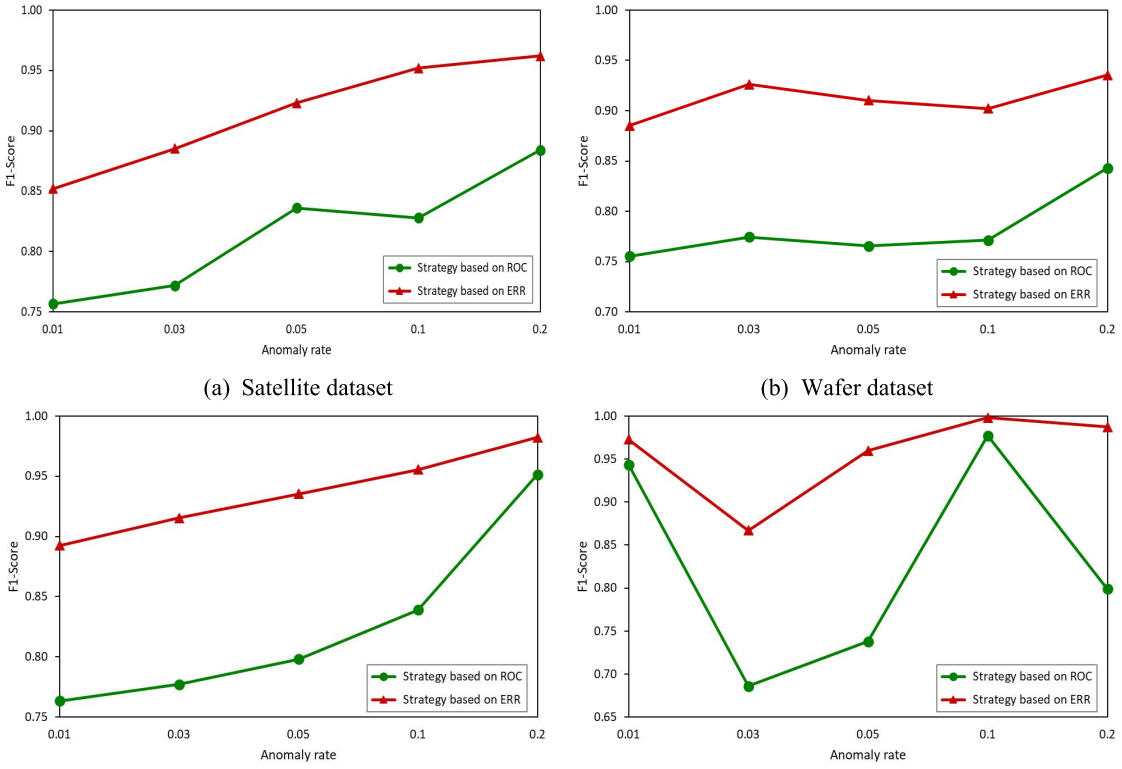
anomaly samples. To combat this problem, we used the two criteria, F1-Score, and G-mean, with different rates of anomalies to evaluate our model's robustness. Fig. 3 shows the chart of F1-Score and G-mean comparisons with different rates of anomalies. The comparison results indicate that the F1-Score and G-mean of most algorithms tend to increase as anomaly rates increase. Our proposed framework's F1-Score and G-mean values are highest on all datasets under different anomaly rates. Therefore, we are justified in concluding that our proposed framework has good robustness even in the face of dataset imbalance problems.

F. THRESHOLD SETTING STRATEGY COMPARISON

In order to verify the effectiveness of the proposed threshold setting strategy, we compared two threshold setting strategies based on four datasets, shown in Table 9, Table 10, Table 11, and Table 12. Table 9, Table 10, Table 11, and Table 12 show the top three thresholds on four datasets. The rank indicates the suitability of the threshold. Therefore, the highest rank is the best threshold. As explained in formulas 14 and 15, the threshold with the lower distance and the ERR will be higher rank. Meanwhile, we also calculate TP, FP, FN, F1, and G-mean corresponding metrics under

TABLE 8. Anomaly detection results of Opt dataset.

Model	Confusion matrix value				Performance evaluation				
	TP	FP	FN	TN	ERR	Precision	Recall	F1-Score	G-mean
KNN	82	23	13	52	0.1465	0.7810	0.8632	0.8200	0.7736
LOF	54	51	32	33	0.3696	0.5143	0.6279	0.5654	0.4967
iForest	78	28	24	40	0.1918	0.7358	0.7647	0.7500	0.6707
OCSVM	80	25	13	52	0.1592	0.7619	0.8602	0.8081	0.7622
GMM	82	23	14	51	0.1474	0.7810	0.8542	0.8159	0.7673
PCA	90	16	12	52	0.1013	0.8491	0.8824	0.8654	0.8214
ABOD	81	24	21	44	0.1611	0.7714	0.7941	0.7826	0.7168
HBOS	77	28	19	46	0.1854	0.7333	0.8021	0.7662	0.7061
CBLOF	82	23	13	52	0.1465	0.7810	0.8632	0.8200	0.7736
MOGAAL	51	54	58	7	0.4821	0.4857	0.4679	0.4766	0.2317
DAGMM	84	22	44	21	0.1732	0.7925	0.6563	0.7179	0.5661
GAN	97	1	6	66	0.0061	0.9898	0.9417	0.9652	0.9166
MAD-GAN	104	25	2	39	0.1488	0.8062	**0.9811**	0.8851	0.6688
OmniAnomaly	105	28	3	34	0.1677	0.7895	0.9722	0.8714	0.6073
Autoencoder	90	9	28	43	0.0634	0.9091	0.7627	0.8295	0.7942
VAEs	89	11	4	66	0.0663	0.8900	0.9570	0.9223	0.9057
LSTM-AE	92	3	35	40	0.0222	0.9684	0.7244	0.8288	0.8209
MSCVAE	99	1	4	66	**0.0060**	**0.9900**	0.9612	**0.9754**	**0.9730**



(a) Satellite dataset (b) Wafer dataset

(c) EEG dataset (d) Opt dataset

FIGURE 4. F1-Score comparisons two threshold setting strategies on four datasets under different anomaly rate.

different thresholds. As stated, the proposed threshold setting strategy can effectively improve the model performance of multivariate time series anomaly detection.

As stated, the ROC-based strategy pays more attention to FN and ignores FP, so it selects the best threshold with lower FN. In contrast, the ERR-based strategy focuses on FP, and its best selected threshold makes for similarly excellent FP performance. Consider the experimental results in Table 9. The first threshold of the ERR-basedstrategy has 4 FP and 6FN, and the corresponding F1-Score and

TABLE 9. Top three thresholds using strategies on satellite dataset.

Strategy	Rank	TP	FP	FN	TN	F1-Score	G-mean
ERR	1	122	4	6	107	0.9606	0.9585
	2	121	3	9	106	0.9528	0.9514
	3	122	7	4	106	0.9569	0.9530
ROC	1	123	28	11	77	0.8632	0.8204
	2	123	32	11	73	0.8512	0.7989
	3	124	35	10	70	0.8464	0.7854

TABLE 10. Top three thresholds using strategies on Wafer dataset.

Strategy	Rank	TP	FP	FN	TN	F1-Score	G-mean
ERR	1	142	2	0	66	0.9930	0.9852
	2	142	3	0	65	0.9895	0.9777
	3	141	2	1	66	0.9895	0.9817
ROC	1	142	2	0	66	0.9930	0.9852
	2	142	3	0	65	0.9895	0.9777
	3	140	2	3	65	0.9825	0.9746

TABLE 11. Top three thresholds using strategies on EEG dataset.

Strategy	Rank	TP	FP	FN	TN	F1-Score	G-mean
ERR	1	296	9	34	151	0.9323	0.9201
	2	293	8	37	152	0.9287	0.9184
	3	292	8	38	152	0.9270	0.9168
ROC	1	297	25	27	141	0.9195	0.8824
	2	296	14	38	142	0.9193	0.8982
	3	297	26	27	140	0.9181	0.8793

TABLE 12. Top three thresholds using strategies on Opt dataset.

Strategy	Rank	TP	FP	FN	TN	F1-Score	G-mean
ERR	1	99	1	4	66	0.9754	0.9730
	2	98	1	5	66	0.9703	0.9681
	3	97	1	6	66	0.9652	0.9632
ROC	1	104	25	2	39	0.8851	0.7732
	2	105	28	3	34	0.8714	0.7302
	3	104	26	2	38	0.8814	0.7632

G-mean are 0.9606 and 0.9585. However, the first threshold of ROC-based strategy has 28 FP and 11 FN, and the corresponding F1-Score and G-mean are just 0.8632 and 0.8204. ROC-based strategy fails to achieve the best threshold because of its excessive attention to FN.

Further investigation was conducted to analyze the same proposed framework using the two threshold setting strategies under different anomaly rates, shown in Fig. 4. The ERR-based strategy achieves a better threshold than the ROC-based strategy under a low anomaly rate on four datasets. Besides, it is clear that the difference between the two strategies decreases as the anomaly rate increases. In short, the proposed threshold setting strategy is superior to the traditional ROC-based strategy and can achieve an appropriate threshold even when confronted with a dataset imbalance problem.

V. CONCLUSION

In this paper, we proposed a novel MSCVAE framework to solve the anomaly detection problem for multivariate time series data. The framework employs multi scale (resolution) system attribute matrices which transform multivariate time series into multi scale attribute matrices. This approach allows for characterizing the state of the entire system in different time segments, and adopts the convolutional variational autoencoder to generate reconstructed attribute matrices, which makes our framework more robust by taking advantage of VAEs. An attention-based Convolutional Long-Short Term Memory (ConvLSTM) network is used to capture the temporal patterns. The framework can model both inter-sensor correlations and temporal dependencies of multivariate time series. Finally, a new ERR-based threshold setting strategy is adopted, instead of a traditional ROC-based threshold setting strategy, to achieve better model performance. To verify the effectiveness of the proposed framework, experiments on four datasets were implemented. The results demonstrate that MSCVAE can outperform state-of-the-art baseline methods.

The work reported here justifies the following conclusions.

(1) Multi-scale attribute matrices provide an effective preprocessing method for characterizing system states at different time segments of multivariate time series with no need of prior knowledge.

(2) CNN structure is embedded in the encoder, and the decoder of the VAEs model is adopted to extract the characteristics of the time series. An attention-based Convolutional Long-Short Term Memory (ConvLSTM) network is used to capture the temporal patterns and reconstruct the attribute matrices, providing an effective unsupervised anomaly detection method. Combined with the proposed ERR-based threshold setting strategy, the MSCVAE based framework can achieve excellent performance.

(3) Experiments on four datasets indicate that our framework is outperforms competing models in detection accuracy and robustness under imbalanced datasets. An extensive experiment was also conducted to verify that the proposed threshold setting strategy can acquire an optimal threshold in the anomaly detection task, thus contributing to the superior anomaly detection performance of our model.

REFERENCES

[1] P. J. Brockwell and R. A. Davis, *Time Series: Theory and Methods*. Cham, Switzerland: Springer, May 2009.

[2] V. Chandola, A. Banerjee, and V. Kumar, "Anomaly detection: A survey," *ACM Comput. Surv.*, vol. 41, no. 3, p. 15, 2009, doi: 10.1145/1541880.1541882.

[3] L. Li, J. Yan, H. Wang, and Y. Jin, "Anomaly detection of time series with smoothness-inducing sequential variational auto-encoder," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 3, pp. 1177–1191, Mar. 2020.

[4] N. Laptev, S. Amizadeh, and I. Flint, "Generic and scalable framework for automated time-series anomaly detection," in *Proc. 21st ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Aug. 2015, pp. 1939–1947, doi: 10.1145/2783258.2788611.

[5] M. Munir, S. A. Siddiqui, A. Dengel, and S. Ahmed, "DeepAnT: A deep learning approach for unsupervised anomaly detection in time series," *IEEE Access*, vol. 7, pp. 1991–2005, 2019, doi: 10.1109/ACCESS.2018.2886457.

[6] H. Izakian and W. Pedrycz, "Anomaly detection and characterization in spatial time series data: A cluster-centric approach," *IEEE Trans. Fuzzy Syst.*, vol. 22, no. 6, pp. 1612–1624, Dec. 2014, doi: 10.1109/TFUZZ.2014.2302456.

[7] R. Chalapathy and S. Chawla, "Deep learning for anomaly detection: A survey," 2019, *arXiv:1901.03407*.

[8] S. Du, T. Li, Y. Yang, and S.-J. Horng, "Multivariate time series forecasting via attention-based encoder–decoder framework," *Neurocomputing*, vol. 388, pp. 269–279, May 2020, doi: 10.1016/j.neucom.2019.12.118.

[9] M. Goldstein and S. Uchida, "A comparative evaluation of unsupervised anomaly detection algorithms for multivariate data," *PLoS ONE*, vol. 11, no. 4, Apr. 2016, Art. no. e0152173, doi: 10.1371/journal.pone.0152173.

[10] J. Li, H. Izakian, W. Pedrycz, and I. Jamal, "Clustering-based anomaly detection in multivariate time series data," *Appl. Soft Comput.*, vol. 100, Mar. 2021, Art. no. 106919, doi: 10.1016/j.asoc.2020.106919.

[11] M. Canizo, I. Triguero, A. Conde, and E. Onieva, "Multi-head CNN-RNN for multi-time series anomaly detection: An industrial case study," *Neurocomputing*, vol. 363, pp. 246–260, Oct. 2019. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0925231219309877

[12] S. Plakias and Y. S. Boutalis, "Exploiting the generative adversarial framework for one-class multi-dimensional fault detection," *Neurocomputing*, vol. 332, pp. 396–405, Mar. 2019, doi: 10.1016/j.neucom.2018.12.041.

[13] K. Hundman, V. Constantinou, C. Laporte, I. Colwell, and T. Soderstrom, "Detecting spacecraft anomalies using LSTMs and nonparametric dynamic thresholding," in *Proc. 24th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Jul. 2018, pp. 387–395, doi: 10.1145/3219819.3219845.

[14] D. Kwon, K. Natarajan, S. C. Suh, H. Kim, and J. Kim, "An empirical study on network anomaly detection using convolutional neural networks," in *Proc. IEEE 38th Int. Conf. Distrib. Comput. Syst. (ICDCS)*, Jul. 2018, pp. 1595–1598.

[15] S. Zavrak and M. Iskefiyeli, "Anomaly-based intrusion detection from network flow features using variational autoencoder," *IEEE Access*, vol. 8, pp. 108346–108358, 2020, doi: 10.1109/ACCESS.2020.3001350.

[16] B. Yan and G. Han, "Effective feature extraction via stacked sparse autoencoder to improve intrusion detection system," *IEEE Access*, vol. 6, pp. 41238–41248, 2018, doi: 10.1109/ACCESS.2018.2858277.

[17] C. Zhang, D. Song, Y. Chen, X. Feng, C. Lumezanu, W. Cheng, J. Ni, B. Zong, H. Chen, and N. V. Chawla, "A deep neural network for unsupervised anomaly detection and diagnosis in multivariate time series data," in *Proc. AAAI Conf. Artif. Intell.*, Jul. 2019, vol. 33, no. 1, pp. 1406–1409, doi: 10.1609/aaai.v33i01.33011409.

[18] H. Liang, L. Song, J. Wang, L. Guo, X. Li, and J. Liang, "Robust unsupervised anomaly detection via multi-time scale DCGANs with forgetting mechanism for industrial multivariate time series," *Neurocomputing*, vol. 423, pp. 444–462, Jan. 2021, doi: 10.1016/j.neucom.2020.10.084.

[19] N. Li and F. Chang, "Video anomaly detection and localization via multivariate Gaussian fully convolution adversarial autoencoder," *Neurocomputing*, vol. 369, pp. 92–105, Dec. 2019, doi: 10.1016/j.neucom.2019.08.044.

[20] T. Chen, X. Liu, B. Xia, W. Wang, and Y. Lai, "Unsupervised anomaly detection of industrial robots using sliding-window convolutional variational autoencoder," *IEEE Access*, vol. 8, pp. 47072–47081, 2020, doi: 10.1109/ACCESS.2020.2977892.

[21] B. Min, J. Yoo, S. Kim, D. Shin, and D. Shin, "Network anomaly detection using memory-augmented deep autoencoder," *IEEE Access*, vol. 9, pp. 104695–104706, 2021, doi: 10.1109/ACCESS.2021.3100087.

[22] S. Naseer, Y. Saleem, S. Khalid, M. K. Bashir, J. Han, M. M. Iqbal, and K. Han, "Enhanced network anomaly detection based on deep neural networks," *IEEE Access*, vol. 6, pp. 48231–48246, 2018, doi: 10.1109/ACCESS.2018.2863036.

[23] A. Lavin and S. Ahmad, "Evaluating real-time anomaly detection algorithms—The numenta anomaly benchmark," in *Proc. IEEE 14th Int. Conf. Mach. Learn. Appl. (ICMLA)*, Dec. 2015, pp. 38–44.

[24] J. Masci, U. Meier, D. C. Cireşan, and J. Schmidhuber, "Stacked convolutional auto-encoders for hierarchical feature extraction," in *Proc. 21st Int. Conf. Artif. Neural Netw.*, Jun. 2011, pp. 52–59.

[25] R. A. A. Habeeb, F. Nasaruddin, A. Gani, I. A. T. Hashem, E. Ahmed, and M. Imran, "Real-time big data processing for anomaly detection: A survey," *Int. J. Inf. Manage.*, vol. 45, pp. 289–307, Apr. 2019, doi: 10.1016/j.ijinfomgt.2018.08.006.

[26] J. Fan, Q. Zhang, J. Zhu, M. Zhang, Z. Yang, and H. Cao, "Robust deep auto-encoding Gaussian process regression for unsupervised anomaly detection," *Neurocomputing*, vol. 376, pp. 180–190, Feb. 2020, doi: 10.1016/j.neucom.2019.09.078.

[27] M. L. Shyu, S. C. Chen, K. Sarinnapakorn, and L. Chang, "A novel anomaly detection scheme based on principal component classifier," Ph.D. dissertation, Dept. Elect. Comput. Eng., Univ. Miami, Coral Gables, FL, USA, 2003.

[28] Y. J. Lee, Y. R. Yeh, and Y. C. F. Wang, "Anomaly detection via online oversampling principal component analysis," *IEEE Trans. Knowl. Data Eng.*, vol. 25, no. 7, pp. 1460–1470, Jul. 2012.

[29] C. C. Aggarwal, "Outlier analysis," in *Data Mining: The Textbook*. Cham, Switzerland: Springer, 2015.

[30] A. Borghesi, A. Bartolini, M. Lombardi, M. Milano, and L. Benini, "Anomaly detection using autoencoders in high performance computing systems," in *Proc. AAAI Conf. Artif. Intell.*, Jul. 2019, vol. 33, no. 1, pp. 9428–9433.

[31] C. Zhou and R. C. Paffenroth, "Anomaly detection with robust deep autoencoders," in *Proc. 23rd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Aug. 2017, pp. 665–674, doi: 10.1145/3097983.3098052.

[32] B. Zong, Q. Song, M. R. Min, W. Cheng, C. Lumezanu, D. Cho, and H. Chen, "Deep autoencoding Gaussian mixture model for unsupervised anomaly detection," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2018, pp. 1–19.

[33] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, "Generative adversarial nets," in *Proc. Adv. Neural Inf. Process. Syst.*, pp. 2672-2680, 2014.

[34] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, 1997.

[35] A. Radford, L. Metz, and S. Chintala, "Unsupervised representation learning with deep convolutional generative adversarial networks," in *Proc. Roc. Int. Conf. Learn. Represent. (ICLR)*, 2016, pp. 1–16. [Online]. Available: http://arxiv.org/pdf/1511.06434.pdf

[36] F. Karim, S. Majumdar, H. Darabi, and S. Chen, "LSTM fully convolutional networks for time series classification," *IEEE Access*, vol. 6, pp. 1662–1669, 2017, doi: 10.1109/ACCESS.2017.2779939.

[37] D. Li, D. Chen, B. Jin, L. Shi, J. Goh, and S. K. Ng, "MAD-GAN: Multivariate anomaly detection for time series data with generative adversarial networks," in *Proc. Int. Conf. Artif. Neural Netw*, 2018, pp. 703–716. [Online]. Available: http://arxiv.org/pdf/1901.04997.pdf

[38] Y. Qin, D. Song, H. Chen, W. Cheng, G. Jiang, and G. W. Cottrell, "A dual-stage attention-based recurrent neural network for time series prediction," in *Proc. 26th Int. Joint Conf. Artif. Intell.*, Jul. 2017, pp. 2627–2633.

[39] Y. Su, Y. Zhao, C. Niu, R. Liu, W. Sun, and D. Pei, "Robust anomaly detection for multivariate time series through stochastic recurrent neural network," in *Proc. 25th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Jul. 2019, pp. 2828–2837, doi: 10.1145/3292500.3330672.

[40] R. Hinami, T. Mei, and S. Satoh, "Joint detection and recounting of abnormal events by learning deep generic knowledge," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 3619–3627.

[41] M. Hu, "Detecting anomalies in time series data via a meta-feature based approach," *IEEE Access*, vol. 6, pp. 27760–27776, 2018, doi: 10.1109/ACCESS.2018.2840086.

[42] M. Hu, X. Feng, Z. Ji, K. Yan, and S. Zhou, "A novel computational approach for discord search with local recurrence rates in multivariate time series," *Inf. Sci.*, vol. 477, pp. 220–233, Mar. 2019.

[43] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, Oct. 2015, pp. 234–241.

[44] D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," in *Proc. Int. Conf. Learn. Represent.*, Apr. 2014, pp. 1–4.

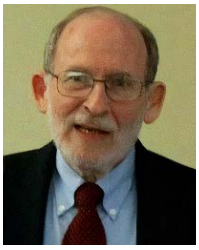
[45] J. Sun, X. Wang, N. Xiong, and J. Shao, "Learning sparse representation with variational auto-encoder for anomaly detection," *IEEE Access*, vol. 6, pp. 33353–33361, 2018, doi: 10.1109/ACCESS.2018.2848210.

[46] X. Wang, Y. Du, S. Lin, P. Cui, Y. Shen, and Y. Yang, "AdVAE: A self-adversarial variational autoencoder with Gaussian anomaly prior knowledge for anomaly detection," *Knowl.-Based Syst.*, vol. 190, Feb. 2020, Art. no. 105187, doi: 10.1016/j.knosys.2019.105187.

[47] R. Yao, C. Liu, L. Zhang, and P. Peng, "Unsupervised anomaly detection using variational auto-encoder based feature extraction," in *Proc. IEEE Int. Conf. Prognostics Health Manage. (ICPHM)*, Jun. 2019, pp. 1–7.
[48] J. An and S. Cho, "Variational autoencoder based anomaly detection using reconstruction probability," *Special Lecture IE*, vol. 2, no. 1, pp. 1–18, Dec. 2015.
[49] J. Bayer and C. Osendorfer, "Learning stochastic recurrent networks," in *Proc. NIPS Workshop Adv. Variational Inf.*, 2014, pp. 1–9.
[50] M. Sölch, J. Bayer, M. Ludersdorfer, and P. van der Smagt, "Variational inference for on-line anomaly detection in high-dimensional time series," in *Proc. ICML*, 2016, pp. 1–4.
[51] D. Park, Y. Hoshi, and C. C. Kemp, "A multimodal anomaly detector for robot-assisted feeding using an LSTM-based variational autoencoder," *IEEE Robot. Automat. Lett.*, vol. 3, no. 3, pp. 1544–1551, Jul. 2018.
[52] J. Pereira and M. Silveira, "Unsupervised anomaly detection in energy time series data using variational recurrent autoencoders with attention," in *Proc. 17th IEEE Int. Conf. Mach. Learn. Appl. (ICMLA)*, Dec. 2018, pp. 1275–1282.
[53] A. Fernández, S. Garcia, M. Galar, C. R. Prati, B. Krawczyk, and F. Herrera, *Learning From Imbalanced Data Sets*. Berlin, Germany: Springer, 2018.
[54] D. Hallac, S. Vare, S. Boyd, and J. Leskovec, "Toeplitz inverse covariance-based clustering of multivariate time series data," in *Proc. 23rd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Aug. 2017, pp. 215–223, doi: 10.1145/3097983.3098060.
[55] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.
[56] S. H. I. Xingjian, Z. Chen, H. Wang, D. Y. Yeung, W. K. Wong, and W. C. Woo, "Convolutional LSTM network: A machine learning approach for precipitation nowcasting," in *Proc. Adv. Neural Inf. Process. Syst.*, 2015, pp. 802–810.
[57] K. Hajian-Tilaki, "Receiver operating characteristic (ROC) curve analysis for medical diagnostic test evaluation," *Caspian J. Internal Med.*, vol. 4, no. 2, pp. 627–635, 2013.
[58] X. Hua, Y. Cheng, H. Wang, Y. Qin, and Y. Li, "Geometric means and medians with applications to target detection," *IET Signal Process.*, vol. 11, no. 6, pp. 711–720, 2017.



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